

Variably strict conditionals

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1. Lewis's analysis, expressed using accessibility and closeness

Following Lewis, let's use ' $A \Box \rightarrow B$ ' to symbolise 'If it were the case that A, it would be the case that B'. Let's also say ' w is an A-world' to mean 'A is true at w ', i.e. 'The proposition expressed by A is true at w '.

New gizmo: a three-place *closeness relation* among possible worlds. Write as ' $u \leq_w v$ ', pronounce as ' u is at least as close to w as v '.

Closeness Analysis: $A \Box \rightarrow B$ is true at w iff either (a) no A-world is accessible from w , or (b) there is an A-world v accessible from w , such B is true at every A-world u such that $u \leq_w v$ and u is accessible from w .

- Lewis only defends this analysis for counterfactuals; he sets indicative conditionals aside. One *could* however defend an analysis of this form for both indicative and counterfactual conditionals; Stalnaker did this at one point. If you do this, you'll need to explain the difference between indicatives and counterfactuals by appeal to some difference in the operative closeness relation and/or accessibility relation.
- When condition (a) holds we say that $A \Box \rightarrow B$ is *vacuously true*; when (b) holds, we say that it is *non-vacuously true*.

2. Closeness and similarity

The Closeness Analysis doesn't say what closeness is: it's a black box that could be filled in any number of different ways. But Lewis has a proposal: for u to be closer to w than v is for u to be *more similar* to w , in relevant respects, than v is.

Objection:

The counterfactual "If Nixon had pressed the button there would have been a nuclear holocaust" is true or can be imagined to be so. Now suppose that there never will be a nuclear holocaust. Then that counterfactual is, on Lewis's analysis, very likely false. For given any world in which antecedent and consequent are both true it will be easier to imagine a closer world in which the antecedent is true and the consequent false. For we need only imagine a change that prevents the holocaust but that does not require such great divergence from actuality. (Kit Fine, *Critical Notice of Counterfactuals*)

Lewis's response:

It is all too easy to make offhand similarity judgments and then assume that they will do for all purposes. But if we respect the extreme shiftiness and context-dependence of similarity, we will not set much store by offhand judgments. We will be prepared to distinguish between the similarity relations that guide our offhand explicit judgments and those that govern our counterfactuals in various contexts....

The presence or absence of a nuclear holocaust surely does contribute with overwhelming weight to some prominent similarity relations.... But the relation that governs the counterfactual may not be one of these. It may nevertheless be a relation of overall similarity ... because it is a resultant, under some system of weights or priorities, of a multitude of relations of similarity in particular respects. (Lewis, 'Counterfactual Dependence and Time's Arrow', pp. 42–3)

3. Accessibility

The notion of “accessibility” does much less work in the Closeness Analysis than in the Strict Conditional Analysis. To see this, consider the following hypothesis:

Universality Hypothesis: every world is accessible from every world

Given the Strict Conditional Analysis, adopting this would have the crazy result that ‘If A, B’ is true only when $A \wedge \neg B$ is metaphysically impossible (true at no worlds). By contrast, the Universality Hypothesis can be (and often is) combined with the Closeness Analysis without leading to any obviously problematic judgments about the truth-values of particular conditionals. If we do this, we can simplify the statement of the Closeness Analysis slightly by dropping all mention of accessibility.

4. The Basic Properties and valid argument-schemes

In order to secure a reasonable logic for conditionals from the Closeness Analysis, we will need to assume that the following “Basic Properties” of the accessibility and closeness relations:

Accessibility is reflexive: every world is accessible from itself.

Transitivity: if $u_1 \leq_w u_2$ and $u_2 \leq_w u_3$ then $u_1 \leq_w u_3$.

Connectedness: for any u , v , and w , either $u \leq_w v$ or $v \leq_w u$.

Weak centering: $w \leq_w v$ for any w and v .

(Note that the connectedness property entails that $v \leq_w v$ for every w and v .)

Provided we require the closeness and accessibility relations to have the Basic Properties, the following argument-schemas will be *valid* (in the sense that no matter how we

interpret the sentence-letters, the conclusion is true at every world where all the premises are true)

ID (Identity): $\vdash A \Box \rightarrow A$

MP (Modus Ponens): $A \Box \rightarrow B, A \vdash B$

CC (Agglomeration): $A \Box \rightarrow B, A \Box \rightarrow C \vdash A \Box \rightarrow (B \wedge C)$

CA (Disjunction): $A \Box \rightarrow C, B \Box \rightarrow C \vdash (A \vee B) \Box \rightarrow C$

CSO (Restricted Transitivity): $A \Box \rightarrow B, B \Box \rightarrow A, A \Box \rightarrow C \vdash B \Box \rightarrow C$

CV (Limited Antecedent Strengthening): $A \Box \rightarrow C \vdash (A \wedge B \Box \rightarrow C) \vee (A \Box \rightarrow \neg B)$

Example 1: To show that MP is valid, suppose that $A \Box \rightarrow B$ and A are both true at some world w . By the reflexivity of accessibility, w is accessible from itself, so A is true at some world accessible from w , so $A \Box \rightarrow B$ is not vacuously true at w . So there is an A -world v accessible from w such that whenever $u \leq_w v$ and u is an A -world accessible from w , B is true at u . By Weak Centering, $w \leq_w v$, and by the reflexivity of accessibility, w is an A -world accessible from w . So B is true at w .

Example 2: To show that CC is valid, suppose that $A \Box \rightarrow B$ and $A \Box \rightarrow C$ are both true at some world w . If there is no A -world accessible from w , then $A \Box \rightarrow (B \wedge C)$ is vacuously true at w . Otherwise, there are A -worlds v_1 and v_2 accessible from w such that for any A -world u accessible from w , B is true at u if $u \leq_w v_1$, and C is true at u if $u \leq_w v_2$. By Connectedness, either $v_1 \leq_w v_2$ or $v_2 \leq_w v_1$. Suppose the former. And suppose that u is an A -world accessible from w such that $u \leq_w v_1$. Then B is true at u ; also $u \leq_w v_2$ by the Transitivity Property, so C is true at u . Therefore $B \wedge C$ is true at u . Since this holds for every u , we can conclude that v_1 is an accessible A -world such that $A \wedge B$ is true at every A -world that is at least as close to w as it. Similarly, if $v_2 \leq_w v_1$, v_2 has the same feature. So either way, $A \Box \rightarrow (B \wedge C)$ is true at w .

We can get further argument-schemas to come out valid by imposing more requirements on the accessibility and/or closeness relations.

One important example: if we require the closeness relation to have the

Strong centering property: For any w and v , $v \leq_w w$ iff $v=w$.

the following schema will be valid

CS (And-to-if): $A, B \vdash A \Box \rightarrow B$

5. Invalidities

Even given the Basic Properties, Antecedent Strengthening is no longer valid: $A \Box \rightarrow C$ can be true at a world without $(A \wedge B) \Box \rightarrow C$ being true at that world.

- We can find a countermodel to this with just two worlds w_1 and w_2 , each accessible from the other, such that $w_2 \leq_{w_1} w_1$ is false. Let A be true at both worlds, B be true only at w_2 , and C be true only at w_1 ; then $A \Box \rightarrow C$ is true at w_1 although $(A \wedge B) \Box \rightarrow C$ is false at w_1 .

Similarly for Transitivity and Contraposition.

6. Reformulation in terms of spheres

(D1) Let a **sphere around** w be a set S of worlds, all accessible from w , such that $u \in S$ whenever u is accessible from w , $u \leq_w v$, and $v \in S$.¹

Notice that given the Basic Properties this definition of ‘sphere’ entails that

- (a) Nestedness: if S_1 and S_2 are both spheres around w , then either $S_1 \subseteq S_2$ or $S_2 \subseteq S_1$.
- (b) The union of any set of spheres around w is a sphere around w
- (c) The intersection of any nonempty set of spheres of around w is a sphere around w .
- (d) Weak centering: Every nonempty sphere around w contains w and there is at least one nonempty sphere around w .

Given (D1), the properties of $\leq$, and the the Closeness Analysis, we can derive

Spheres Analysis: $A \Box \rightarrow C$ is true at w iff either (a) no sphere around w contains an A -world, or (b) some sphere around w contains an A -world and is such that every A -world in it is a B -world.

Alternatively, we could take the notion of a sphere around a world as primitive, and define accessibility and closeness in terms of it, as follows:

(D2) v is accessible from $w =_{df}$ v belongs to some sphere around w .

(D3) $u \leq_w v =_{df}$ every sphere around w that contains v also contains u .

Note that given the assumption that spheres obey conditions (a)–(d), this definition entails that accessibility and closeness have the Basic Properties.

Given (D2) and (D3), conditions (a)–(d) on spheres, and the Spheres Analysis, we can derive the Closeness Analysis. Hence the two formulations are completely equivalent.

¹ The version of this handout distributed in class erroneously omitted the requirement that u be accessible from w . Thanks to Alex Poste for spotting the error.